

Day - 3: REST APIs and HTTP Methods

What are REST APIs?

- **REST (Representational State Transfer)** is a **type of API** that follows **specific rules and guidelines** for communication.
 - It uses **HTTP methods** like:
 - GET
 - POST
 - PUT
 - DELETE
 - It defines **fixed standards** for how requests and responses should be structured.
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Ways to Send Data in Requests

1. req.body

- Hidden in the request payload.
- Best for **larger, sensitive, or complex** data (like passwords).
- Typically used in **POST**, **PUT**, and **PATCH** requests.

2. req.query

- Data sent in the URL after **?** as key-value pairs.
- Ideal for **small, optional parameters**.
- Avoid using it for **sensitive or complex data**.
- Example:

```
GET /search?gender=ma le&age=24
```

3. req.params

- Data sent as part of the API path.
- Commonly used for identifying **specific resources**.
- Example:

```
GET /user/ankur_bit_io
```

- Accessed in code as:

```
req.params.username
```



REST API Methods

Method	Purpose
GET	Retrieve data from the server.
POST	Send new data to the server (create).
PATCH	Update existing data on the server.
DELETE	Remove/delete data from the server.

💡 Example Use Case:

A social media app like `x.com` may use:

- `req.params` to fetch user profiles (`/user/:username`)
- `req.query` to filter search results (`/search?gender=male`)
- `req.body` for login forms or post creation